Q-Learning

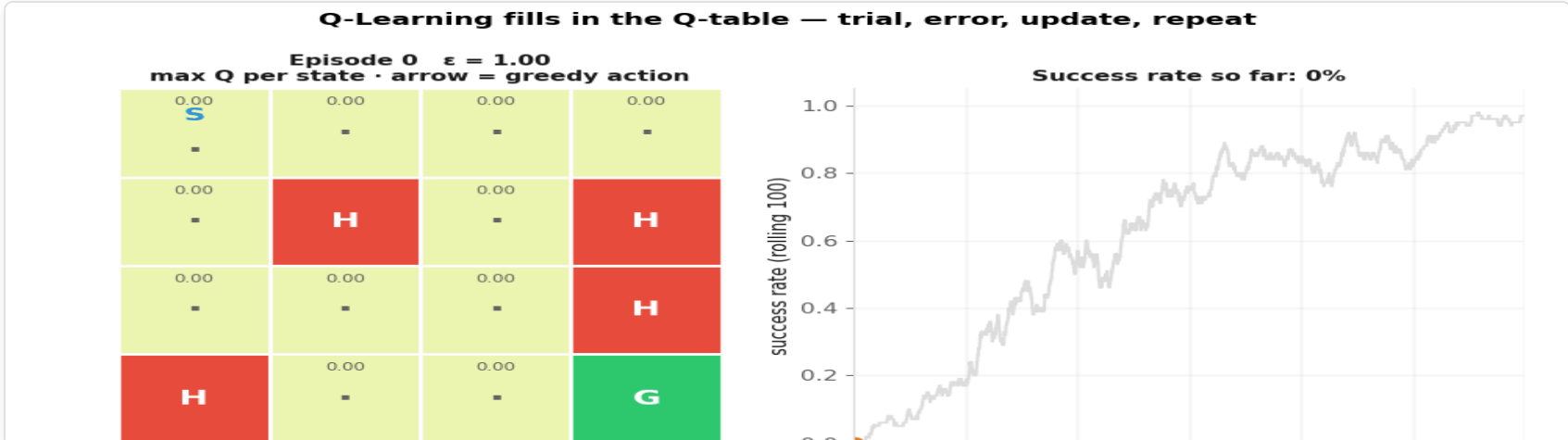
Applied Machine Learning — Session 4, Chapter 2

From Ch10 to Ch11

Ch10: if we *know* Q(s, a) for every state and action, the policy is trivial: `argmax`.

Ch11: how do we **learn** Q from experience — without knowing the map, the transition probabilities or the reward function?

Answer: start with Q = 0 everywhere, act (ε-greedy), and after *every step* nudge Q(s, a) toward what we just observed.



The Q-Table

```
Q-table (16 states × 4 actions) — all zeros at the start
           ←      ↓      →      ↑
state 0:   0.00   0.00   0.00   0.00
state 1:   0.00   0.00   0.00   0.00
...
state 14:  0.00   0.00   0.00   0.00
state 15:  0.00   0.00   0.00   0.00   ← goal (never left)
```

- One number per (state, action): *"how good is it to do a in s?"*
- **Policy** = `argmax` per row
- **Learning** = updating these numbers from experience

The TD Update (1/2): the idea

After one step (s, a) → r, s' we have **new information**:

```
what we thought:   Q(s, a)
what we just saw:  r  +  γ · max_a' Q(s', a')          ← "target"
                   ^        ^
              reward now     best we believe we can still get from s'
```

Move the estimate a little toward the target:

```
Q(s, a)  ←  Q(s, a)  +  α · ( target - Q(s, a) )
                             └──── TD error ─────┘
```

α = learning rate (how big a step). If the episode ended at s', there is no future: target = r.

The TD Update (2/2): a worked step

GridWorld, α = 0.1, γ = 0.99, Q all zeros. The agent stands in state 14 and steps → into the goal:

```
target  = 1.0 + (episode ended → no future) = 1.0
TD err  = 1.0 - 0.0 = 1.0
Q(14,→) = 0.0 + 0.1 · 1.0 = 0.10
```

Next episode: from state 13 it steps → into 14 (reward −0.01):

```
target  = -0.01 + 0.99 · max Q(14, ·) = -0.01 + 0.99 · 0.10 = 0.089
Q(13,→) = 0.0 + 0.1 · (0.089 - 0.0)   = 0.0089
```

→ Value **flows backwards** from the goal, one cell per successful visit. Small numbers, but the *ordering* of actions is what matters for `argmax`.

The Full Algorithm

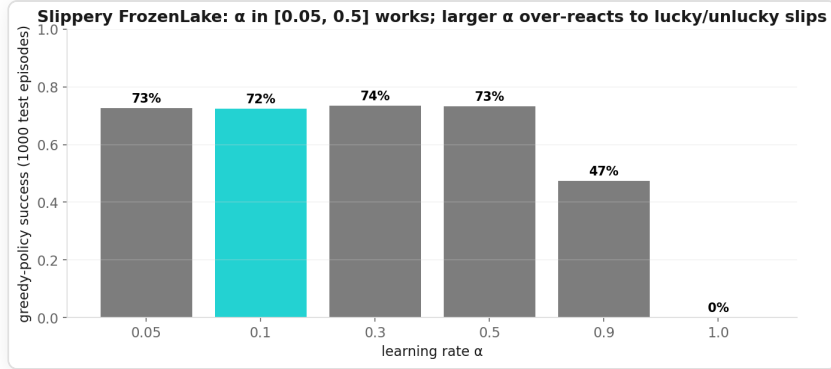
```
Q, epsilon = np.zeros((n_states, n_actions)), 1.0
for episode in range(n_episodes):
    state = env_reset()
    for step in range(max_steps):
        # ε-greedy
        if rng.random() < epsilon:  action = rng.integers(n_actions)      # explore
        else:                       action = np.argmax(Q[state])          # exploit

        next_state, reward, done = env_step(state, action)   # Gymnasium: obs, r, terminated, trunc
        target = reward + (0 if done else gamma * np.max(Q[next_state]))  # TD target
        Q[state, action] += alpha * (target - Q[state, action])           # TD update

        state = next_state
        if done: break
    epsilon = max(epsilon * 0.999, 0.01)                                  # decay
```

Hyperparameters (the ones we use)

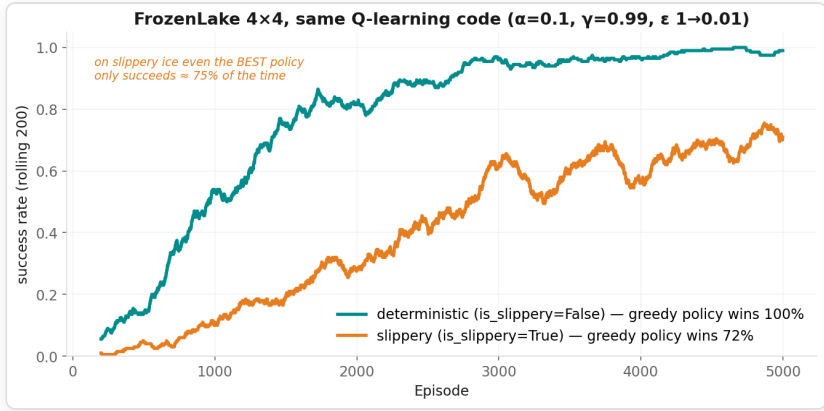
Parameter	Symbol	Ours	Typical
learning rate	α	**0.1**	0.05 – 0.5
discount	γ	**0.99**	0.9 – 0.99
exploration	ε	**1 → 0.01**	×0.999 / episode
episodes		**3 000 – 5 000**	10^3 – 10^5



- α: step size of each update
- γ: how far value flows backwards
- more episodes = better, slower

FrozenLake (Gymnasium) — same map, two difficulty levels

```
SFFF       S start (0)
FHFH       F frozen
FFFH       H hole → episode ends, reward 0
HFFG       G goal (15) → reward +1
```



- 16 states, 4 actions (←0 ↓1 →2 ↑3)
- reward **only** at the goal → sparse signal
- `is_slippery=True` : intended move with p = 1/3, else sideways

Same code, two worlds: deterministic → ~100 %; slippery → ~70 %. On slippery ice even the *optimal* policy only wins ≈ 3 of 4 episodes.

Why is RL hard? (and why does it still work?)

Sparse rewards — FrozenLake: one +1 at the very end, zeros everywhere else. Random exploration must stumble into the goal first.

Stochasticity — same state, same action, different outcome. Estimates need many samples (→ small α).

Credit assignment — which of the 20 earlier moves caused the success? Answer: γ propagates value backwards, one step per success.

It still works because: tabular Q-learning provably converges to Q^* if every (s, a) is visited infinitely often and α decreases suitably. In practice: enough episodes + ε-decay.

Quick Check

1. α = 1 in a deterministic world. What happens? And on slippery ice?

2. γ = 0. Which Q-values ever become non-zero in FrozenLake (reward only at the goal)?

3. After training, ε is still 0.05. Is the "success rate during training" the success rate of the *learned policy*?

Quick Check

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- **1.** Deterministic: fine — the target is exact, so overwrite it. Slippery: each slip overwrites the estimate; Q jumps around and never settles (see the α bar chart: 0 % at α = 1).
- **2.** Only entries whose step lands *in* the goal — essentially Q(14, →) (on slippery ice also slips from 14). Nothing flows backwards → the agent cannot find the goal from the start.
- **3.** No — 5 % of the actions are random. Evaluate the greedy policy separately (`evaluate_greedy` in the demo).

Now: Live Demo (~8 min)

→ `02-examples/ch11_rl_algorithms_examples.ipynb`

1. Wrap Gymnasium's FrozenLake in our `reset / step` interface
2. Q-Learning in ~25 lines — train on solid ice
3. **Same code** on slippery ice → compare success rates
4. Q-table heatmaps + greedy policy arrows: read what the agent learned

Now: Exercise (~10 min)

→ `03-exercises/ch11_rl_algorithms_exercises.ipynb`

Tasks 1–4 (10 min): Q-table → ε-greedy → TD update → fill three lines in the given training loop. Each task has a ▶ check cell with the expected numbers.

Bonus A: run *your* loop on slippery FrozenLake (Gymnasium) — why not 100 %? **Bonus B:** SARSA — change one line, compare learning curves.

Further Reading (not needed for the capstone)

Idea	One line	Where
SARSA (on-policy)	target uses the action you *actually* take next: `r + γ Q(s', a')` — learns the value of the ε-greedy behaviour → more cautious near cliffs	Bonus B
Deep Q-Network (DQN)	replace the table by a neural network Q(s, a; θ) — Atari from pixels (2013)	
Policy gradient / PPO	learn π(a|s) directly; needed for continuous actions (robot joints); PPO is the workhorse behind RLHF	
Model-based RL	learn P and R, then plan (AlphaZero uses a known model + search)	
Off- vs on-policy	Q-learning learns the *greedy* policy while behaving ε-greedy (off-policy); SARSA learns the policy it follows (on-policy)	

Key Takeaways

- Q-Learning: **Q-table + ε-greedy + TD update**, repeated over many episodes
- Target `r + γ · max Q(s', ·)` ; TD error = target − estimate; move by α
- Value flows **backwards** from the reward — that solves credit assignment
- Same code works on deterministic and stochastic worlds; the **environment** sets the ceiling
- Evaluate the **greedy** policy, not the training run (train ≠ test — again)

Next: Chapter 12

Capstone: End-to-End ML Workflow

"Time to put everything together."